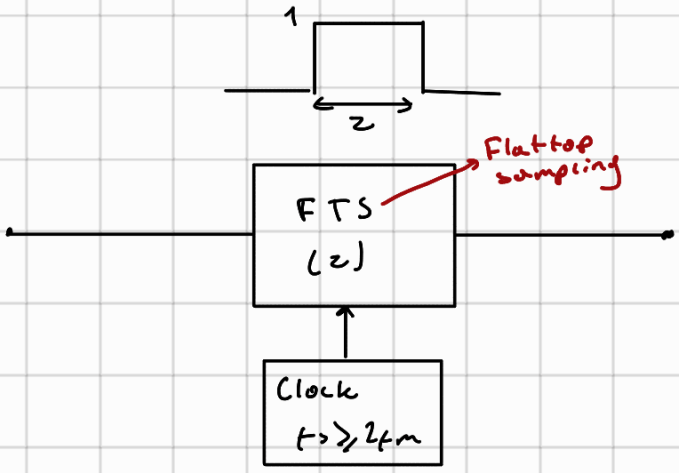
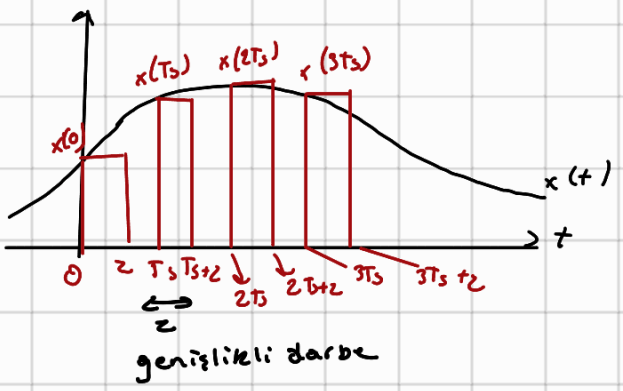
Sample and Hold operation (Örnekle vetut işlemi)

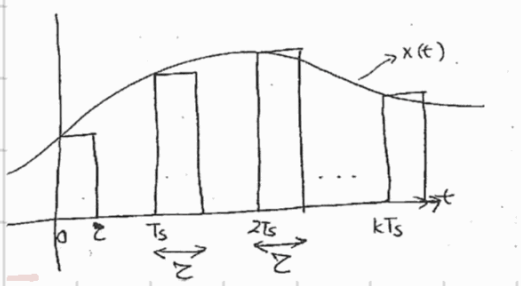
Düz Tepeli örnekleme
(Flat top sampling) (FTS)

* tepeyi takip etmiyor





Sample and holol circuit
(Örnekle ve tut devresi)



$x_s(t)$: Darbe üzerinden modüle (DÜM)

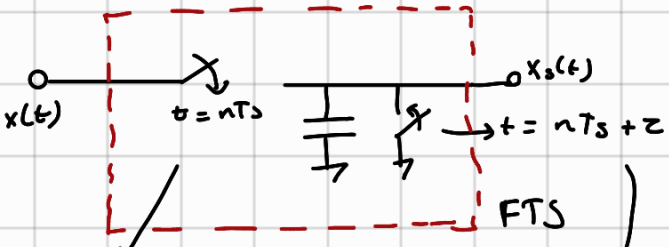
Pulse amplitude mod

Burda darbeler düz tepeli

bu discrete time analog modulation zaman ayrık ama genlik değil

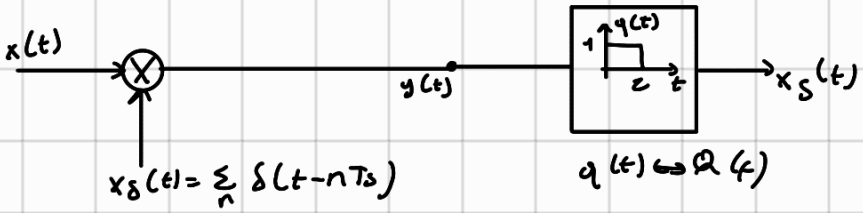
Desiscrete time analog modulation

Sample / Hold (Damp) circuit



nTs anında hemen kapanıp açılıyor

τ saniye sonra bu anahtar kapanıyor ve gerilim sıfır oluyor



$$y(t) = x(t)\, x_\delta(t) = \sum_n x(nT_s)\, \delta(t-nT_s)$$

$$x_s(t) = q(t) \circledast y(t) = \sum_n x(nT_s)\, q(t-nT_s)$$

$$x_s(t) = q(t) \circledast [x(t)\, x_\delta(t)] = q(t) \circledast \left[\sum_n x(nT_s)\, \delta(t-nT_s)\right] = \sum_n x(nT_s)\, q(t-nT_s)$$

$$X_s(f) = Q(f)\left[X(f) \circledast X_\delta(f)\right] = Q(f)\left[X(f) \circledast f_s \sum_n \delta(f-nf_s)\right]$$

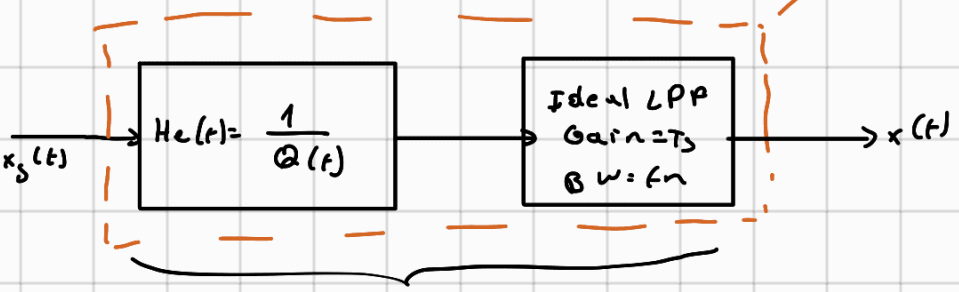
$$X_s(f) = Q(f)\left[f_s \sum_n X(f-nf_s)\right]$$

↳ $z\,\mathrm{sinc}(fz)\,e^{-j2\pi f \frac{z}{2}}$

$$X_s(f) = f_s\, Q(f) X(f) + Q(f) \cdot f_s \sum_{n\neq 0} X(f-nf_s)$$

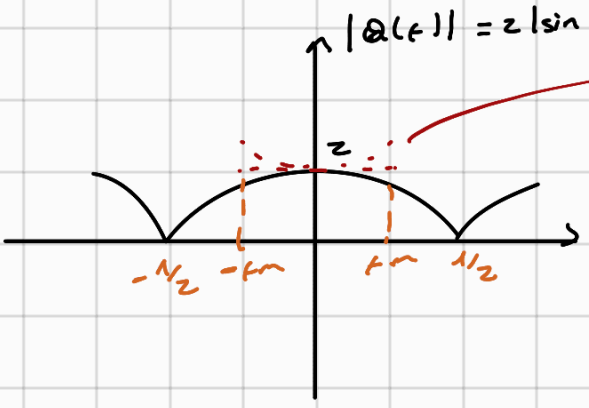
!!

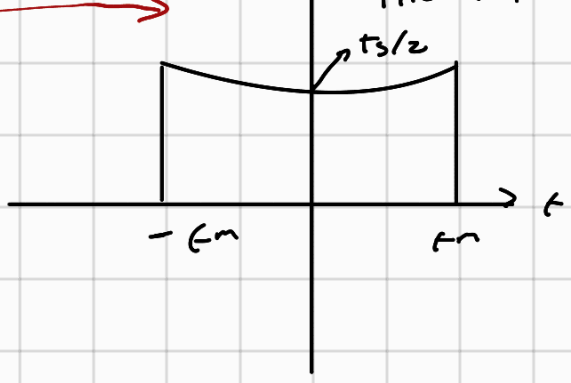




$$H_{eo}(f) = \begin{cases} \frac{T_s}{Q(f)} & |f| \le f_m \\ 0 & \text{else} \end{cases}$$

$$H_{eo}(f) = \begin{cases} \frac{T_s}{Q(f)} & |f| \le f_m \\ 0 & \text{else} \end{cases}$$





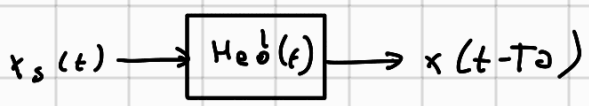
$$H_{eo}(f) = \frac{T_s}{Q(f)} = \frac{T_s}{z\,\mathrm{sinc}(fz)}\, \underbrace{e^{j2\pi f \frac{z}{2}}}_{\text{advance in time} \rightarrow h_{eo}(t) \text{ is not causal}}, \quad |f| \le f_m$$

For this reason, $H_{eo}'(f) = H_{eo}(f)\, e^{-j2\pi f T_x}$ is chosen

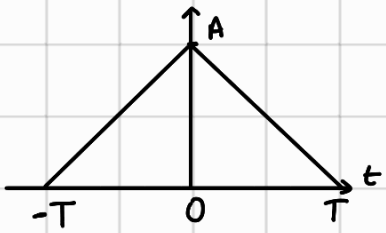
↓

$$h_{eo}'(t) = h_{eo}(t - T_x)$$

$T_x = \frac{z}{2} + T_0$

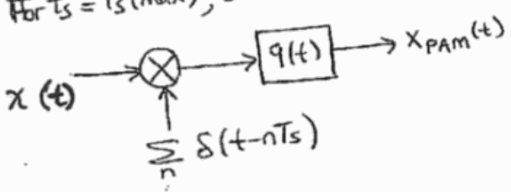


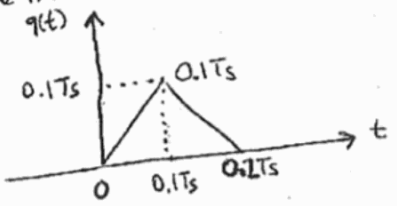
Reminder



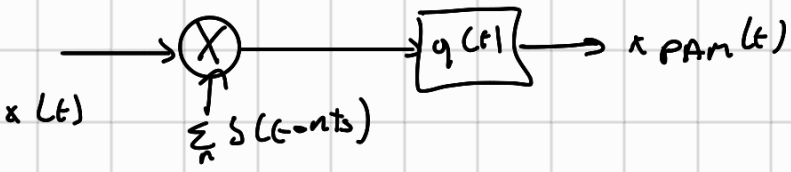
$$A\wedge\left(\frac{t}{T}\right)\xleftrightarrow{\mathcal{F}} AT\,\mathrm{sinc}^2(fT)$$

Ex. (2004 midterm Ex. Q.)
A PAM signal is produced as shown in the figure below. a) If the bandwidth of $x(t)$ is 1 kHz, what is the maximum value of T_s to obtain $x(t)$ without distortion at the receiver?
b) For $T_s = T_s(max)$, sketch PAM demodulator. Find the transfer function of demodulator.





$0.1T_s\,\mathrm{sinc}^2(0.1T_s f)\,e^{-j2\pi f\,0.1T_s}$

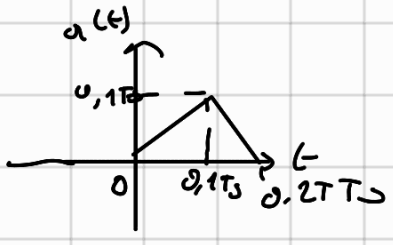


a) $BW\{x(t)\} = 1\,kHz = f_m$

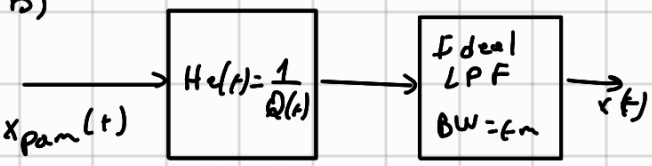
$f_s \geq 2f_m = 2kHz$

$T_s \leq \frac{1}{2f_m} = 0,5\,msec$

$T_s(max) = 0,5\,msec$



b)

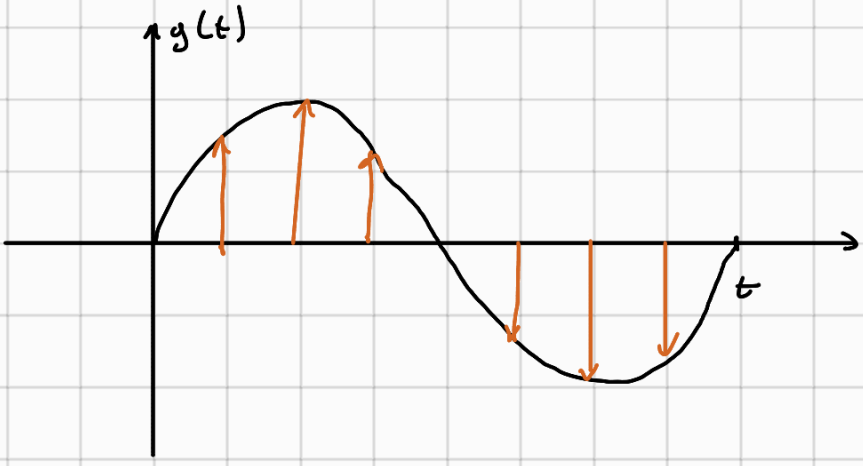


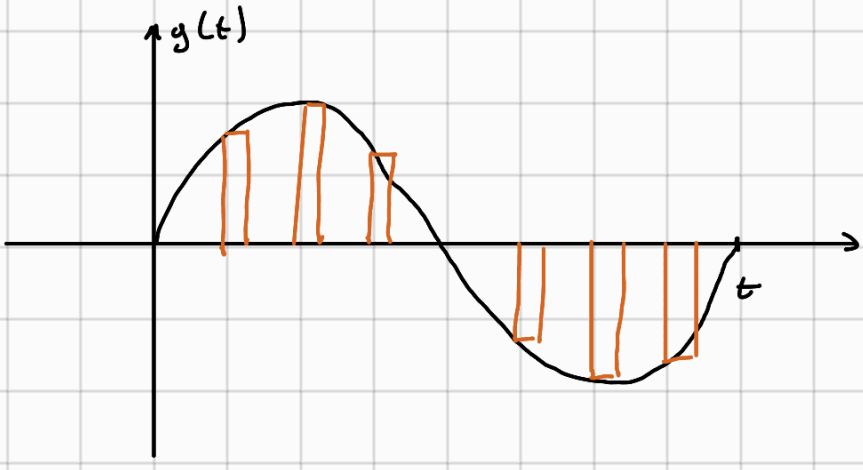
$$Q(f) = 0,1T_s\left(0,1T_s\,\mathrm{sinc}^2(0,1T_s f)\,e^{-j2\pi f\,0,1T_s}\right)$$
$$= 0,01\,T_s^2\,\mathrm{sinc}^2(0,1T_s f)\,e^{-j2\pi f\,0,1T_s}$$

$$H_{eo}(f) = \begin{cases} \frac{T_s}{Q(f)}, & |f| \leq f_m \\ 0, & |f| > f_m \end{cases}$$

$$H_{eo}(f) = \frac{T_s\,e^{j2\pi f\,0,1T_s}}{0,01\,T_s^2\,\mathrm{sinc}^2(0,1T_s f)}$$

$$= \begin{cases} \frac{2\cdot10^5\,e^{j2\pi f\,5\cdot10^{-5}}}{\mathrm{sinc}^2(5\cdot10^{-5}f)}, & |f| \leq 1\,kHz \text{ iken} \\ 0, & \text{else} \end{cases}$$

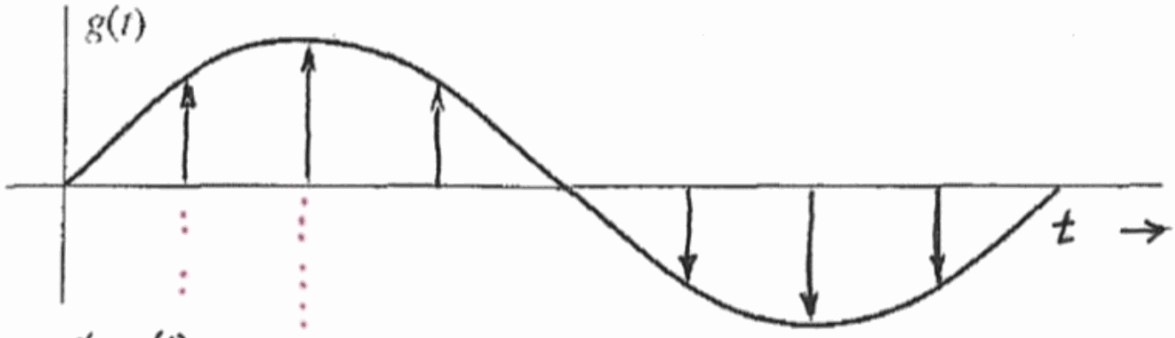


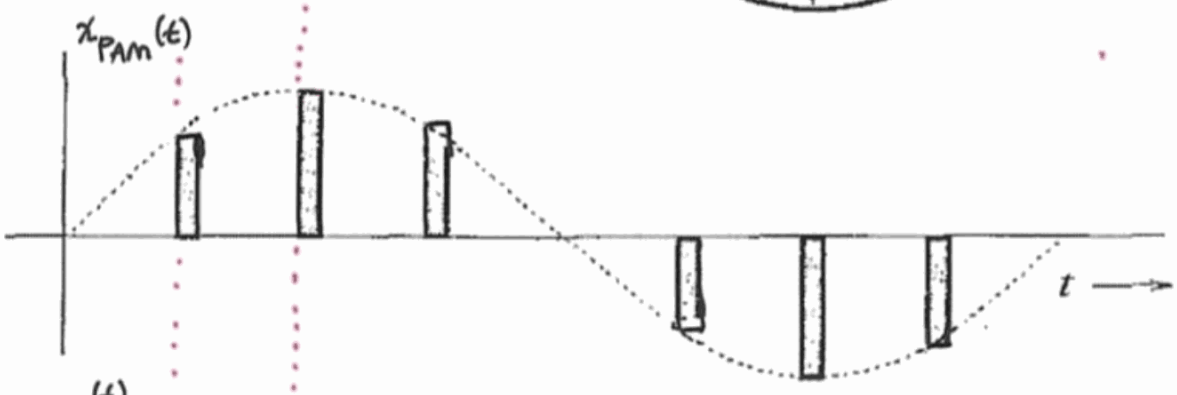






xpp~[t] : Pulse Width modulation



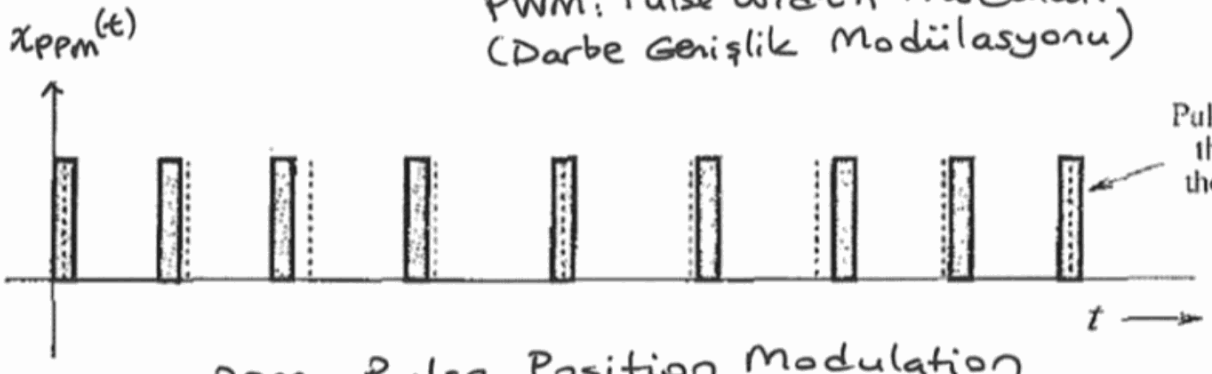


PAM



PWM

PWM: Pulse Width Modulation
(Darbe Genişlik Modülasyonu)

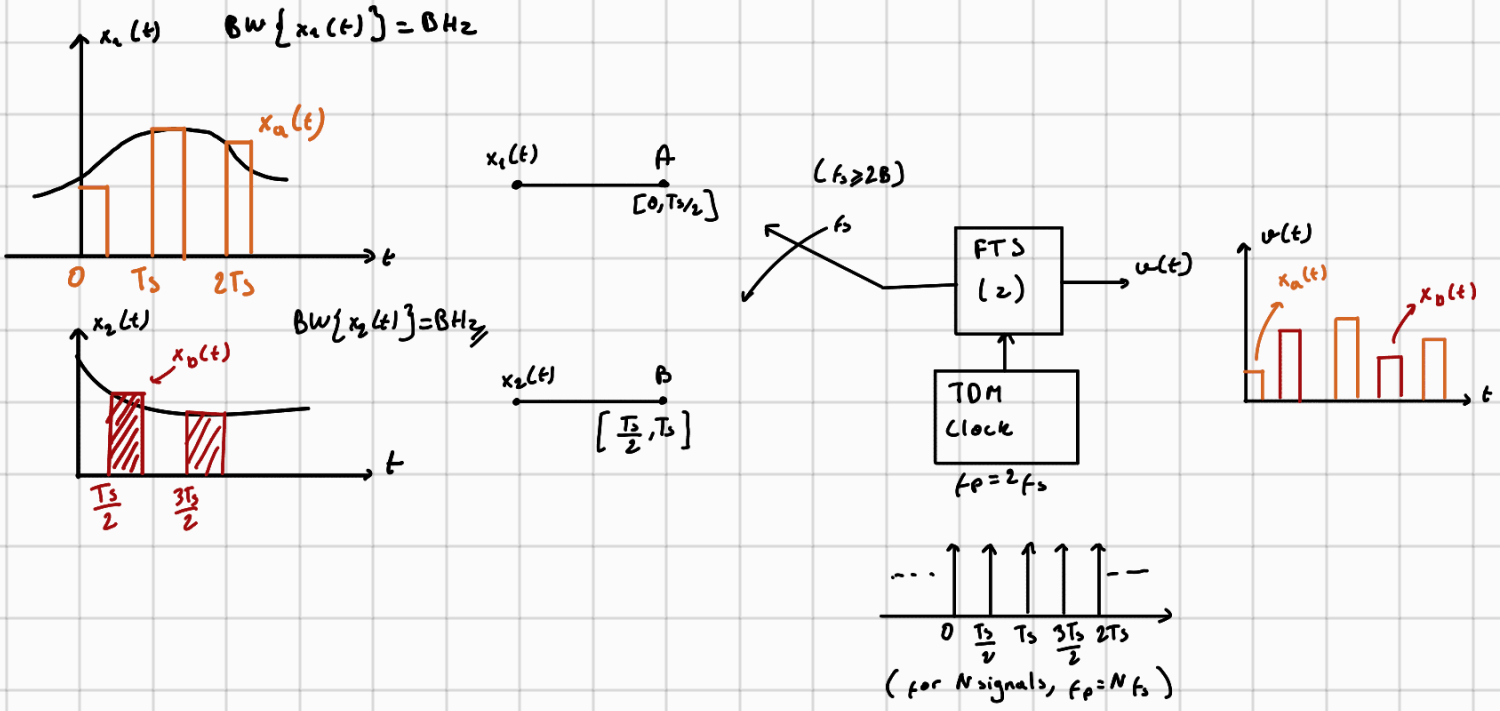


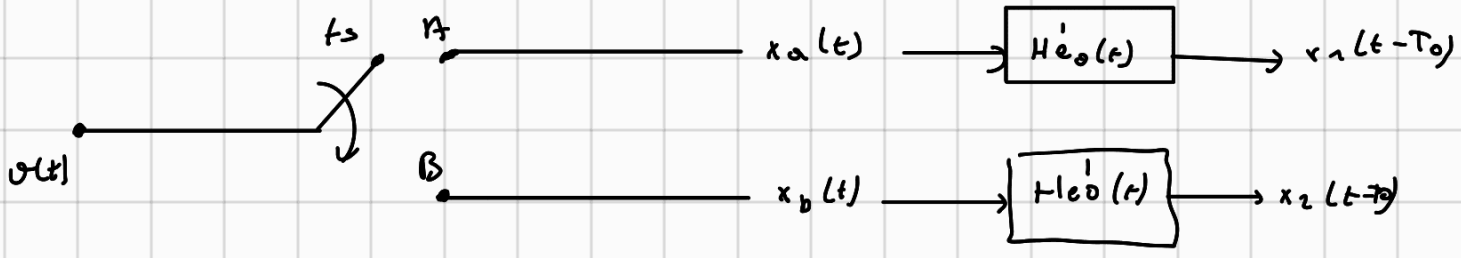
PPM

PPM: Pulse Position Modulation
(Darbe Pozisyon Modülasyonu)

Time division multiplexing (TDM) — Zaman bölmeli çoklama (ZBÇ)

- Multiple PAM signals are transmitted without overlapping.
- At the receiver, PAM signals are seperated by synchronization.





FDM x TDM

Advantages: 1) Easier to implement 2) Less crosstalk than FDM

Disavantages: 1) Strict synchronization

Çok kaynaklı (Multi-source) → multiplexing (çoklama)

Çok kullanıcılı (Multi user) → Multiple Access (çoklu erişim)

↳ TDMA, FDMA

2. nesil avrupadada 3G

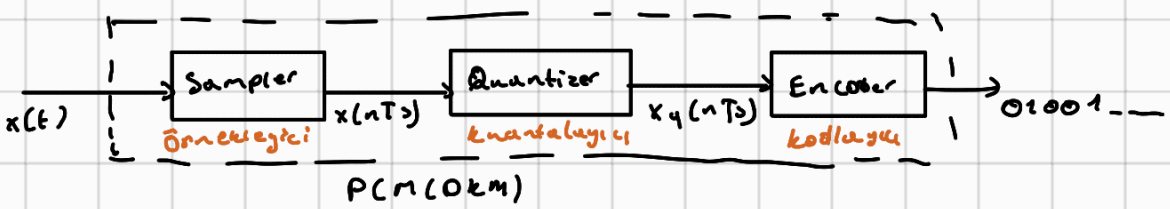
1. nesil

3. nesil Code Devision multiplicing

4. ne sil OFTM

Pulse code Modulation

(Darbe kod modülasyonu - DKM)



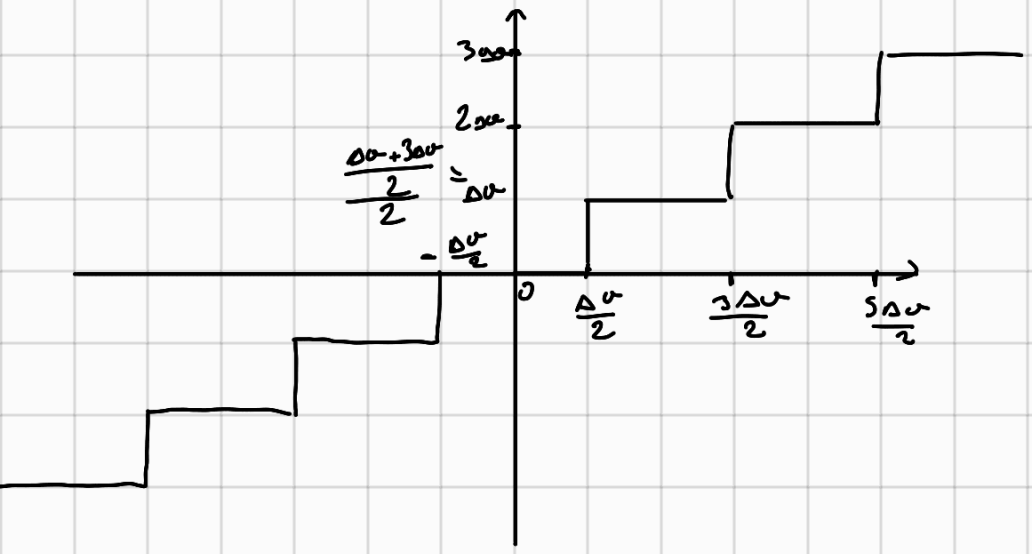
x(t) analog signal

$x_p = maks|x(t)|$ peak value (Tepe değeri)

$-x_p$ den $+x_p$ ye kadar

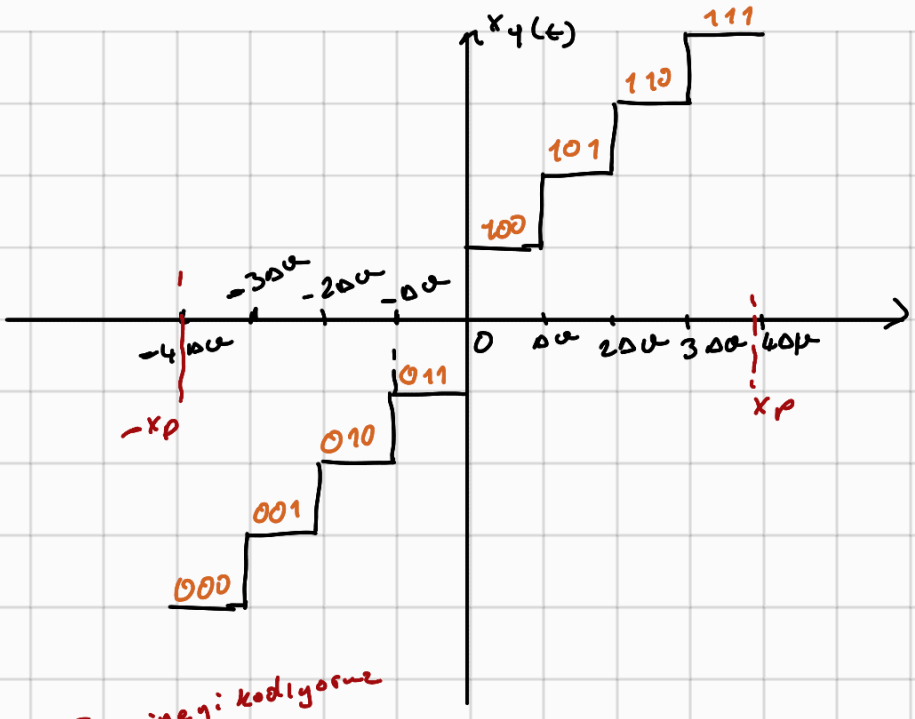
kuantalayıcı adımı = $\Delta v = \frac{2x_p}{L}$ → kuantalayıcıdaki seviye sayısı (No of levels in quantizer)

(Quantizer step)



Assume $\Delta v = 1V$

x(t)		$x_q(t)$
1,07	→	1V
1,40V	→	1V
0,55V	→	1V
2,25V	→	2V
-3,1 V	→	-3V



8 seviyeli kodlayorsun

$L = 8$

(Mid rise quantizer)

Assume $\Delta v = 1V$

$x(t)$	$x_q(t)$
1,03V	1,5V
0,55V	0,5V
-3,1V	-3,5V

1,01V → 1,5V $+\frac{\Delta v}{2}$ hata

$-\frac{\Delta v}{2}$ hata

* Each sample is approximated with a quantized value, the approximation will result in an error less than $\Delta v/2$ in the positive direction or $-\Delta v/2$ in the negative "

Quantization error : $(-\Delta v/2, \Delta v/2)$

$x_q(t) = x(t) + q(t)$ → kuantulama gürültüsü (Quantization noise)

↳ Analog işaret

↳ kuantulanmış işaret

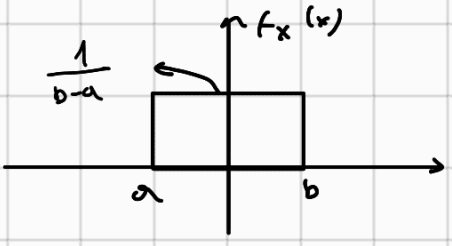
q is uniform in $[-\Delta v/2, \Delta v/2]$

Average quantization noise power = $N_q = E[q^2]$

(ort. kuantalama gürültüsü gücü)

Reminder

X RD $[a, b]$ düzgün d.



$\mu_x = \frac{a+b}{2}$

$\sigma_x^2 = \frac{(b-a)^2}{12}$

$\sigma_q^2 = E[(q-m_q)^2] = E[q^2] = N_q$

$N_q = \frac{[\Delta v/2 - (-\Delta v/2)]^2}{12} = \frac{\Delta v^2}{12}$

$\Delta v = \frac{2x_p}{L} \Rightarrow N_q = \frac{\Delta v^2}{12} = \frac{x_p^2}{3L^2}$

İşaretin gücü = $S_0 = \langle x^2(t) \rangle = P_x$

Gürültünün gücü : $N_0 = N_q$

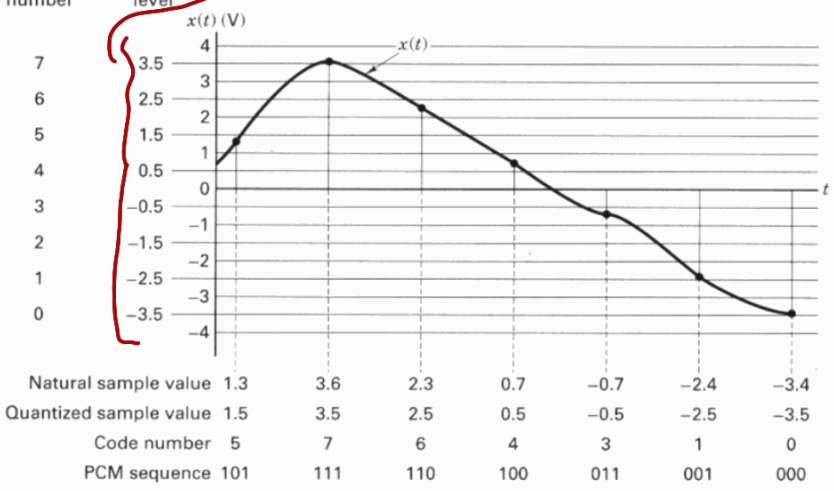
İşaretin gürültüye oranı (igo)

Signal to noise Ratio (SNR)

Signal to Quantization noise Ratio (SQNR)

$$SNR = \frac{S_0}{N_0} = \frac{P_x}{x_p^2} \cdot 3L^2$$

Quantization intervals



Natural sample value	1.3	3.6	2.3	0.7	−0.7	−2.4	−3.4
Quantized sample value	1.5	3.5	2.5	0.5	−0.5	−2.5	−3.5
Code number	5	7	6	4	3	1	0
PCM sequence	101	111	110	100	011	001	000

Figure 2.16 Natural samples, quantized samples, and pulse code modulation. (Reprinted with permission from Taub and Schilling, *Principles of Communications Systems*, McGraw-Hill Book Company, New York, 1971, Fig. 6.5-1, p. 205.)

* Each sample is quantized to the middle of the interval